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Experiment Name: Z-Test,T-Test,F-Test Anova

TEST T TEST F
TEST ANOVA

In [1]:
```
import numpy as np
from scipy import stats
```

In [3]:
```
marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])
mu_0 = 70
```

In [5]:
```
t_stat, p_value = stats.ttest_1samp(marks, mu_0)
print(f"T-statistic: {t_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha = 0.05
if p_value < alpha:
   print("Reject Null Hypothesis → Mean is significantly different from 70."
else:
   print("Fail to Reject Null Hypothesis → No significant difference.")
```

```
T-statistic: 1.993
P-value: 0.0774
Fail to Reject Null Hypothesis → No significant difference.
```

In [6]:
```
import numpy as np
from math import sqrt
from scipy.stats import norm
```

In [7]:
```
x_bar = 51.2
mu_0 = 50
sigma = 3
n = 36
```

In [8]:
```
z_stat = (x_bar - mu_0) / (sigma / sqrt(n))
p_value = 2 * (1 - norm.cdf(abs(z_stat)))
print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha = 0.05
if p_value < alpha:
   print("Reject Null Hypothesis → Mean is significantly different from 50 g
else:
   print("Fail to Reject Null Hypothesis → No significant difference.")
```

```
Z-statistic: 2.400
P-value: 0.0164
Reject Null Hypothesis → Mean is significantly different from 50 g.
```

In [9]:
```
import numpy as np
from scipy import stats
```

In [10]:
```
A = [20, 22, 23]
B = [19, 20, 18]
C = [25, 27, 26]
```

In [12]:
```
f_stat, p_value = stats.f_oneway(A, B, C)

print(f"F-statistic: {f_stat:.3f}")
print(f"P-value: {p_value:.4f}")
```

```
alpha = 0.05
if p_value < alpha:
  print("Reject Null Hypothesis → Means are significantly different.")
else:
  print("Fail to Reject Null Hypothesis → No significant difference.")
```

```
F-statistic: 25.923
P-value: 0.0011
Reject Null Hypothesis → Means are significantly different.
```

In [ ]: